

Computer Programming I COSC-1550

FA 2026 Section 50 3 Credits 09/14/2026 to 12/18/2026 Modified 07/25/2026

Concourse provides instructors a place to build a complete electronic course syllabus with important course information such as institutional policies already in place. Students can ask the course instructor for any information related to this course not available here - note that syllabi may remain undeveloped until closer to the beginning of the course. If you are the course instructor, [click here to learn more about building your syllabus \(https://worldclassroom.webster.edu/courses/1346121/pages/concourse-syllabus-introduction?module_item_id=14473157\)](https://worldclassroom.webster.edu/courses/1346121/pages/concourse-syllabus-introduction?module_item_id=14473157).

Meeting Times

Lecture

Tuesday, Thursday, 10:00 AM to 11:20 AM, EAB 104

Contact Information

Instructor: Dr. Xiaoyuan Suo

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Office: EAB371

Phone: 3142462359

Office Hours

Monday, Tuesday, Wednesday, Thursday, 1:00 PM to 3:00 PM, EAB 371

Description

This course introduces students to the C++ language in order to teach programming as a systematic discipline and as a problem-solving tool. Acquaints students with fundamental concepts of computers, information processing, algorithms and programs. Only offered in a 16-week format. May be repeated once for credit.

Outcomes

At the completion of this course this student will be able to:

1. Write short programs in pseudocode and in the C++ programming language.
2. Write programs that require: making decisions (if and if..else statements), writing loops (while, do..while, and for loops), using functions and using arrays.
3. Read a simple C++ program and predict the outcome of the program.



Required Textbook

C++ from control structures through Objects

Author: Gaddis

Publisher: Pearson

Edition: 10th

ISBN: 978-0134895734



Deliverables



Evaluation

Criteria

programming assignments	10%	All programs must be submitted and run correctly to pass the course All programs must execute according to specifications If an assignment is submitted late, 20% is deducted from the grade, with an additional 2% per day deduction, thereafter.
Attendance (homework and participation)	5%	Attendance and participation are required. Students who cannot attend every class are advised to withdraw from the course. Please notify the instructor as soon as possible regarding an excused absence. A student's final grade will be lowered for each unexcused absence.
Chapter test (total of 5)	70%	14% each, test1: chapter 1, 2, and 3 test2: chapter 4 test3: chapter5 test4: chapter6 test5: chapter 7
Final Exam	15%	Final Exam is comprehensive

lowest % for an A is 93%, lowest % for an A- is 90%, lowest % for an B+ is 87%, lowest % for an B is 83%, lowest % for an B- is 80%, lowest % for an C+ is 77%, lowest % for an C is 73%, lowest % for an C- is 70%, lowest % for an D+ is 67%, lowest% for an D is 63%, any lower grade earns a Fail

There will be 6 Chapter tests.

The UNDERGRADUATE catalog provides these guidelines and grading options:

A, A- superior work in the opinion of the instructor

B+, B, B- good work in the opinion of the instructor

C+, C, C- satisfactory work in the opinion of the instructor

D+, D passing, but less than satisfactory work in the opinion of the instructor
I incomplete work in the opinion of

the instructor

ZF An incomplete which was not completed within one year of the end of the course

F unsatisfactory work in the opinion of the instructor; no credit is granted

W withdrawn from the course

IP course in progress

NR not reported for the course

Z a temporary designation given by the registrar indicating that the final grade has not been submitted by the

instructor. When the final grade is filed in the Office of the Registrar, that grade will replace the Z.

Schedule

C++ CS1 16-Week Course Calendar - Webster University

Week	Event	Content	Supporting
1	Getting Started with C++	Course introduction; computer systems; C++ development environment; basic program structure; output	Compile, run, and simple C++ programs
2	Basic Calculations and Console Input/Output	Variables; constants; data types; arithmetic expressions; input and output	Emphasize expression types, and programs
4 and 5	Decision-Making Program	Decision structures; relational and logical operators; Boolean expressions	Use if, if/else, and decisions
6 and 7	Loop-Based Problem Solving	Loops; repetition structures; validation; counters and accumulators	Use while, do-while loops

6 and 7	Sequential vs. Parallel Sum Demonstration	OpenMP fundamentals; compiler directives; data parallelism; reduction; timing and benchmarking	Compare runtime why small inputs speedup
8 and 9	Modular Program Using Functions	Functions; parameters; return values; scope; program decomposition	Connect function decomposition to decomposition
10	File Processing Assignment	File input/output; processing structured data from files	Read, process, and file-based data
11 and 12	Array Processing and Linear Search	Arrays; array traversal; searching and counting	Prepare for parallel data partitioning
11 and 12	PDC Concept Activities: Coin Search, Flag Coloring, and Parallel Search Animation	Introduction to PDC; sequential vs. parallel execution; task decomposition; workload distribution; animations and unplugged activities	Use short visualizations; guided discussion approximately 30 total
13	2D Array or Grid Simulation	Two-dimensional arrays; nested loops; grid-based simulations	Prepare students for Zombie Attack assignment
14 and 15	Odd-Even Parallel Sort Demo and Zombie Attack Assignment	Parallel algorithms; odd-even sort; synchronization; performance trade-offs; domain decomposition and scalability	Zombie Attack review earlier and continue Week 15
16	Final Exam	final exam review and final exam	Covers C++ fundamentals problem solving, and introductory

* Course Policies

Classtime Behavior

- Since this class is only 50 minutes in length, please do not leave the classroom during the class time to get a snack, to get a drink, to use the restroom, etc., unless it is a true emergency.
- No cell phone usage during class time, including sending or receiving of phone calls, text messages, or instant messages. Phones should be turned off, not set to vibrate, etc. Phones should be not on the desktop or in your lap.
- No laptop computers, tablets, or other electronic devices may be used during class time. All electronic devices should be turned off and should not be on the desktop during class time.
- Students are expected to treat classmates with respect. No student may mock or intimidate a classmate who is struggling or makes a mistake.
- Please use the restroom prior to the start of each quarter exam. You may not leave the room during the exam, unless it is a true emergency.

Attendance and Participation

- Attendance is required. Students who cannot attend every class are advised to withdraw from the course. Please notify the instructor as soon as possible regarding an excused absence. A student's Participation/Attendance grade may be lowered for each unexcused absence.
- Please arrive on time for each class session.
- Class participation is required. Every student is expected to make a positive contribution to each class.
- Students are responsible for making arrangements with other students for missed class notes and with the instructor for handouts and assignments.
- Students should make arrangements with the instructor beforehand if not able to attend any exam. Exams taken late will have late points deducted. Any unexcused absence will result in a failing grade for that exam.
- When required by the University (due to bad weather or other issues) to have a class using Zoom, students should login to the Zoom class session with video and audio.

Exams

- All exams will be conducted in person on campus. Once you begin an exam, please refrain from leaving the room until you are prepared to submit your completed exam.
- Make-up exams must be pre-arranged with the instructor before the exam date, supported by acceptable evidence such as doctor's notes.
- Exam dates will be announced in class with at least a 2-day notice.
- All exams are the property of the department and must be returned to the instructor at the conclusion of the exam.

Homework

- A small homework will be due before class time.
- Late homework is generally not accepted.
- Homework is usually short, and meant to reinforce what was covered in class or meant to assess your ability to solve new problems by extending what was covered in class.
- I do not expect perfection on the homework. I do expect a valid attempt achieved by your own efforts without cheating, copying or "looking" for the solution online or in resource materials.
- Any submitted homework that appears to be copied from classmates, ChatGPT, online or from other materials will result in a failing grade for that assignment.
- Please do not work with other students and turn in "group solutions". All assignments should be completed independently.

Programming Assignments

- All programming assignments must meet the Standard Department Program Style Guidelines.
- All programming assignments must compile and run correctly (according to directions given in lecture and specifications stated on green assignment sheet) upon submission.
- Every assigned program must be submitted and run correctly (according to specifications) in order to pass the course. (Any missing programming assignment will result in a failing grade for the

course.)

- All programming assignments must be your individual work. Any code copied from another student will be considered cheating.
- Do not resubmit a graded programming assignment unless requested by the instructor.

Late Programming Assignments

- Late programming assignments will get significantly lower grades.
- If a programming assignment is submitted late, 20% will be deducted from the grade, with an additional 2% per day deduction thereafter.
- Any programming assignment that is submitted that does not compile or that obviously does not meet specifications, 35 - 50% will be deducted from the grade, with an additional 2% per day deduction thereafter until a correct version is submitted.
 - Do not submit a program that does not compile or run correctly. You may see me for help during office hours.

Academic Honesty

- Students are encouraged to discuss concepts and algorithms of the projects together but must code each program independently.
 - Just because you are nice and your classmate begs you, is not a good reason to share your code. You will have the same consequences as the person who does the copying.
- Academic Dishonesty: Submitting an assignment that was copied, whole or in part, from another student (past or present) constitutes a violation of student conduct and will be handled according to the guidelines of the Student Handbook.
- Facilitating Academic Dishonesty: Assisting any person commit an act of academic dishonesty also constitutes a violation of student conduct and will be handled according to the guidelines of the Student Handbook.
- Use of AI: **No AI on coursework.** You may not use any AI tools for any graded work—homework, assignments, projects, quizzes/exams, or discussion posts/replies—unless I explicitly permit AI in writing for that specific task.
 - “AI tools” include (but aren’t limited to) generative assistants and features such as ChatGPT/Copilot/Gemini/Claude; AI search or “assist” modes; AI code completions (e.g., Copilot, CodeWhisperer); debuggers/explainers; paraphrasers/re-writers (e.g., QuillBot); translation tools that propose wording; citation/summary generators; image/audio generators; and AI embedded in LMS, IDEs, browsers, or apps (e.g., Grammarly’s generative features, Bing/Google AI answers). Prohibited uses include brainstorming/outlining, translating/paraphrasing prompts or drafts, generating/editing/checking answers or code, finding bugs, or “just checking if I’m right.”
 - You may not copy, adapt, or be “influenced by” AI output, nor ask anyone else to use AI on your behalf; uploading course materials or your work to AI systems is also prohibited. Allowed supports: textbook/lecture materials, your notes, human help explicitly permitted by the assignment, calculators if allowed, non-AI editor features (syntax highlighting, bracket matching), and basic spelling/typo correction that does not suggest phrasing.

- **It's okay to be wrong**—ask me questions and show your thinking. I may verify authorship by asking you to explain or reproduce parts of your work; inability to do so may be treated as evidence of unauthorized assistance. Violations will be handled as academic dishonesty (e.g., zero on the work, course penalties, and referral under university policy).

Department Policies

- Quizzes, Tests and Exams are the property of the Computer Science Department.
- This syllabus may be revised at the discretion of the instructor without the prior notification or consent of the student.

Institutional Policies

Academic Policies

Academic policies provide students with important rights and responsibilities. Students are expected to familiarize themselves with all academic policies that apply to them. Academic policies for undergraduate students can be found in the Undergraduate Studies Catalog; graduate students should review the Graduate Studies Catalog.

Undergraduate Studies Catalog

The Undergraduate Studies Catalog contains academic policies that apply to all undergraduate students. The *academic policies and information* section of the catalog contains important information related to **attendance, conduct, academic honesty, grades, and more**. If you are an undergraduate student, please review the catalog each academic year. The current Undergraduate Studies Catalog is at:

<http://www.webster.edu/catalog/current/undergraduate-catalog/>
(<http://www.webster.edu/catalog/current/undergraduate-catalog/>).

Graduate Studies Catalog

The Graduate Studies Catalog contains academic policies that apply to all graduate students. The *academic policies* section of the catalog contains important information related to **conduct, academic honesty, grades, and more**. If you are a graduate student, please review the catalog each academic year. The current Graduate Studies Catalog is at:

<http://www.webster.edu/catalog/current/graduate-catalog/>
(<http://www.webster.edu/catalog/current/graduate-catalog/>).

Grading

The Grades section of the academic catalog outlines the various grading systems courses may use, including the information about the final grade reported for this class.

Undergraduate

<http://www.webster.edu/catalog/current/undergraduate-catalog/academic-policies.html#grading>
(<http://www.webster.edu/catalog/current/undergraduate-catalog/academic-policies.html#grading>).

Graduate

<https://webster.edu/catalog/current/graduate-catalog/academic-policies.html#gradingsystem>
(<https://webster.edu/catalog/current/graduate-catalog/academic-policies.html#gradingsystem>).

Incomplete

There are important policies that govern grades of Incomplete (I), including the circumstances under which Incomplete grades are granted, deadlines for completion, and consequences should the remaining course work not be completed. It is the responsibility of a student who requests an Incomplete to ensure that they understand and follow the policies.

Grade Appeals

Instructors are responsible for assigning grades, and student should discuss grade issues with the instructor. Policies and procedures for appealing grades are available in the appropriate catalog.

Academic Honesty Policy

Webster University is committed to academic excellence. As part of our Statement of Ethics, we strive to preserve academic honor and integrity by repudiating all forms of academic and intellectual dishonesty, including cheating, plagiarism and all other forms of academic dishonesty. Academic dishonesty is unacceptable and is subject to a disciplinary response. Students are encouraged to talk to instructors about any questions they may have regarding how to properly credit others' work, including paraphrasing, quoting, and citation formatting. The university reserves the right to utilize electronic databases, such as Turnitin.com, to assist faculty and students with their academic work.

The university's Academic Honesty Policy is published in academic catalogs:

Undergraduate

<http://www.webster.edu/catalog/current/undergraduate-catalog/academic-policies.html>
(<http://www.webster.edu/catalog/current/undergraduate-catalog/academic-policies.html>).

Graduate

<http://www.webster.edu/catalog/current/graduate-catalog/academic-policies.html>
(<http://www.webster.edu/catalog/current/graduate-catalog/academic-policies.html>).

As a part of the university commitment to academic excellence, the Reeg Academic Resource Center provides student resources to become better acquainted with academic honesty and the tools to prevent plagiarism in its many forms:

<https://websteru.sharepoint.com/sites/AcademicIntegrityAtWebster>
(<https://websteru.sharepoint.com/sites/AcademicIntegrityAtWebster>).

Statement of Ethics

Webster University strives to be a center of academic excellence. The university makes every effort to ensure the following:

- The opportunity for students to learn and inquire freely
- The protection of intellectual freedom and the rights of professors to teach
- The advancement of knowledge through scholarly pursuits and relevant dialogue

To review Webster University's statement of ethics, see the Undergraduate Studies Catalog and the Graduate and Studies Catalog:

Undergraduate

<http://www.webster.edu/catalog/current/undergraduate-catalog/overview.html#ethics>
(<http://www.webster.edu/catalog/current/undergraduate-catalog/overview.html#ethics>).

Graduate

<http://www.webster.edu/catalog/current/graduate-catalog/ethics.html>
(<http://www.webster.edu/catalog/current/graduate-catalog/ethics.html>).

Contact Hours for this Course

It is essential that all classes meet for the full instructional time as scheduled. A class cannot be shortened in length. If a class session is cancelled for any reason, the content must be covered at another time.

Important Academic Resources

Academic Accommodations and Support

At Webster University, we understand that all students learn differently and may have specific, often differing, learning needs. Some examples are doing rather than watching, being active instead of sitting still and taking frequent breaks. For others, the professor needs to speak louder, give more clear instructions, and be understanding with health issues. Regardless of what makes learning easier for you, our goal is to provide the most accessible learning environment possible for each student. If there are any specific needs you'd like to make your instructors personally aware of, including any academic accommodations, please reach out to them directly.

Webster University supports students in several ways. Students with academic/learning, health, physical, and psychological disabilities and students who have accessibility concerns can reach out to Webster's Academic ADA Coordinator at disability@webster.edu. You can also reach out if you have any questions relating to accessibility.

Online resources through WorldClassRoom (Canvas) have a variety of accessibility tools, provided through a program called Ally, for both students and faculty. Ally is free and already incorporated into WorldClassRoom. It helps students who have visibility and attention challenges; who need easier

access to course materials from mobile phones, tablets, electronic readers, etc.; or who need or prefer file types different from the format uploaded by the instructor. You can find more information about Ally at <https://webster.edu/online/ally.php> (<https://webster.edu/online/ally.php>).

Life circumstances can make focusing on education, or anything else, difficult. For this reason, several additional resources and services are available to students at no cost:

1. Reeg Academic Resource Center (<https://websteru.sharepoint.com/sites/ReegARC> (<https://websteru.sharepoint.com/sites/ReegARC>))
 - Academic Counseling: academiccounseling@webster.edu
 - Assistive Technology: atacr@webster.edu
 - Disability Accommodations: disability@webster.edu
 - Testing Center: arc@webster.edu
 - Tutoring: tutoring@webster.edu
 - Writing services: writing@webster.edu
2. Counseling and Life Development (<https://www.webster.edu/student-counseling.php> (<https://www.webster.edu/student-counseling.php>)): 314-968–7030 or counselingld@webster.edu (these contact methods are non-confidential, but appointments are confidential)

Lastly, circumstances change. If, at any point throughout your education, you find that something is making learning difficult for you, please reach out to your instructor directly and/or to Webster's ADA team. It is never too late to ask for accommodations. We entirely support you and want you to have the best and most accessible education possible.

Student Success Portal

Webster University's Student Success Portal, powered by Starfish, is a communications tool to connect students with faculty members and campus support services. It allows faculty and staff members to communicate with you regarding academic achievements as well as areas where support may be helpful. You may use the portal to contact faculty and staff members for assistance and to arrange meetings. Activity in the portal will be communicated to your Webster University email account. The Student Success Portal is available via your course home page in WorldClassRoom or via Connections. Learn more about the Student Success Portal at <https://www.webster.edu/success/index.php> (<https://www.webster.edu/success/index.php>).

University Library

Webster University Library is dedicated to supporting the research needs and intellectual pursuits of students throughout the university's worldwide network. Resources include print and electronic books, journal articles, online databases, DVDs and streaming video, CDs and streaming music, datasets, and other specialized information. Services include providing materials at no cost and research help for basic questions to in-depth exploration of resources. The gateway to all of these resources and services is <http://library.webster.edu> (<http://library.webster.edu/>). For support navigating the library's resources, see <http://libanswers.webster.edu/> (<http://libanswers.webster.edu/>), for the many ways to contact library staff.

Drops and Withdrawals

Drop and withdrawal policies dictate processes for students who wish to unenroll from a course. Students must take proactive steps to unenroll; informing the instructor is not sufficient, nor is failing to attend. In the early days of the term or semester, students may DROP a course with no notation on their student record. After the DROP deadline, students may WITHDRAW from a course; in the case of a WITHDRAW, a grade of W appears on the student record. After the WITHDRAW deadline, students may not unenroll from a course. Policies and a calendar of deadlines for DROP and WITHDRAW are at:

Undergraduate

<http://www.webster.edu/catalog/current/undergraduate-catalog/enrollment.html>
(<http://www.webster.edu/catalog/current/undergraduate-catalog/enrollment.html>).

Graduate

<http://www.webster.edu/catalog/current/graduate-catalog/enrollment.html>
(<http://www.webster.edu/catalog/current/graduate-catalog/enrollment.html>).

Academic Calendar - <http://www.webster.edu/academics/academic-calendar/>
(<http://www.webster.edu/academics/academic-calendar/>).

Current tuition rates, policies, and procedures, including details of pro-rated tuition refunds, are available in the “Tuition, Fees, and Refunds” section of Webster’s Academic Catalogs:

Undergraduate

<http://www.webster.edu/catalog/current/undergraduate-catalog/tuition.html>
(<http://www.webster.edu/catalog/current/undergraduate-catalog/tuition.html>).

Graduate

<http://www.webster.edu/catalog/current/graduate-catalog/tuition.html>
(<http://www.webster.edu/catalog/current/graduate-catalog/tuition.html>).

Student Handbook and Other Important Policies

Student handbook and other non-academic policies may apply to you and may impact your experience in this class. Such policies include the student code of conduct, privacy, technology and communications, and more. Please review the handbook each year and be aware of policies that apply to you. The handbook is available at:

<http://www.webster.edu/student-handbook/> (<http://www.webster.edu/student-handbook/>).

Sexual Assault, Harassment, and Other Sexual Offenses

Webster University makes every effort to educate the community to prevent sexual assault, harassment, and other sexual offenses from occurring, and is committed to providing support to those affected when this behavior does occur. To access information and resources or to review the Policy on Sexual Assault, Harassment, and Other Sexual Offenses, visit:

<https://webster.edu/titleix/policy.php> (<https://webster.edu/titleix/policy.php>).

Research on Human Subjects

The Webster University Institutional Review Committee (IRB) is responsible for the review of all research on human subjects. The IRB process applies to all Webster University faculty, staff, and students and must be completed prior to any contact with human subjects. For more information on the IRB, visit:

<https://websteru.sharepoint.com/sites/InstitutionalReviewBoard>
(<https://nam04.safelinks.protection.outlook.com/?url=https%3A%2F%2Fwebsteru.sharepoint.com%2Fsites%2FInstitutionalReviewBoard&data=04%7C01%7Celvirm31%40webster.edu%7C3cc448f960d5427acd08d997d92f31%7Cf57ccca6fb064b328226baf224ff198d%7C0%7C0%7C637707781316471159%7CUnknown%7CTWFpbGZsb3d8eyJWljoImC4wLjAwMDAiLCJQIjoiV2luMzliLCJBTil6Ik1haWwiLCJXVCI6Mn0%3D%7C1000&sdata=YaeRiduWQUA9NOzcmOw4wbrKu46RKKNPd3dokHblcY%3D&reserved=0>).

Student Experience Feedback

At the end of this course, you will have the opportunity to provide feedback about your experience. Your input is extremely valuable to the university, your instructor, and the department that offers this course. Please provide your honest and thoughtful feedback, as it helps the university to provide the best experience possible for all of its students.

Important Technology Information

Connections Accounts

Webster University provides all students, faculty, and staff with a University email account through Connections. Students are expected to activate their Connections account and regularly check incoming university emails. Students may choose to have their University email forwarded to an alternate email address. Connections account holders can call the Help Desk (314-246-5995 or toll-free at 1-866-435-7270) for assistance with this setup. Instructions are also provided on the Information Technology website at:

<https://www.webster.edu/technology/index.php> (<https://www.webster.edu/technology/index.php>).

WorldClassRoom

WorldClassRoom is Webster's Learning Content Management System (LMS). Your instructor may use WorldClassRoom to deliver important information, hold class activities, communicate grades and feedback, and more. WorldClassRoom is available using your Connections ID at:

<https://worldclassroom.webster.edu/> (<https://worldclassroom.webster.edu/>).

Webster Alerts

Webster Alerts is the university's preferred emergency mass notification service, available free to current students, faculty, and staff at all US campuses. By registering a valid cell phone number and email address, you will receive urgent campus text, voice mail, and email communications. Valuable information concerning a range of incidents affecting you - from weather-related campus closures, class delays and cancellations, to more serious or life-threatening events - are immediately and simultaneously delivered through multiple communication channels. To register for Webster Alerts, visit (sign with your university-provided email and password:

<https://www.getrave.com/login/webster> (<https://www.getrave.com/login/webster>).

Campus Information
